

Well logs and [sequence stratigraphic interpretation](#)

The associated pages are focused on well logs and building [sequences](#) stratigraphic models for both

- Clastic and
- [Carbonate depositional systems](#).

The sections are organized as follows:-

Well Log Interpretation

[Introduction to Well Log Sequence Stratigraphy](#)

[Well Log Suites](#)

[Response Character](#)

[Stacking patterns](#)

[High Frequency Clastic Sequence Stratigraphy Venezuela](#)

[Guarico Electric Logs](#)

[High Frequency Carbonate Sequence Stratigraphy](#)

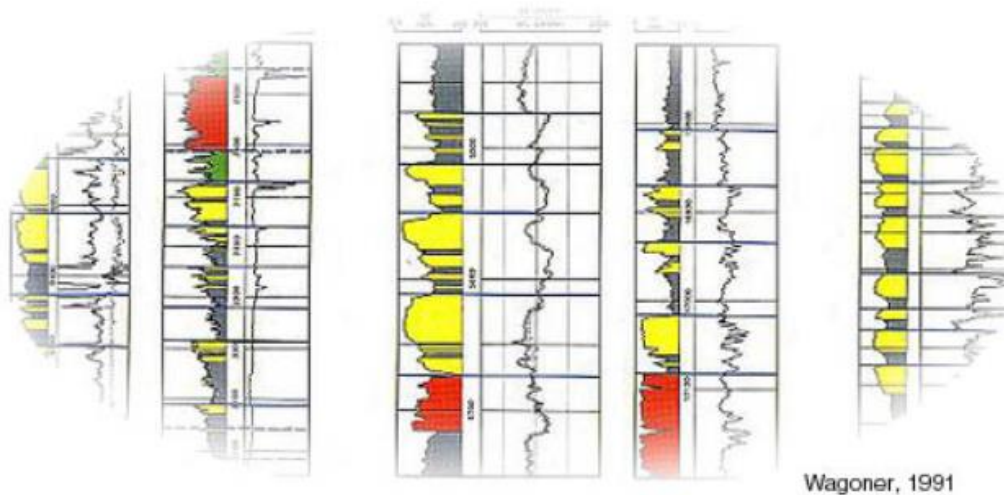
[Hanifa Fm Exercise](#)

[Permian Basin Exercise](#)

[Skydavian Exercise](#)

[References](#)

Use of well logs for [sequence](#) stratigraphic interpretation of the subsurface



The following pages are focused on the use of well logs to build [sequences](#) stratigraphic models of [depositional systems](#). For this process well logs are primarily used to establish the [grain size](#) and the [lithology](#) of the penetrated sediment.

The best [sequence](#) stratigraphic models of the sedimentary fill of [basins](#) are provided by a combination of seismic data, well logs and cores and outcrop studies in conjunction with [biostratigraphy](#). The cores and well logs and outcrop studies provide access to a detailed vertical resolution of sedimentary sections while seismic and outcrop studies provide the [lateral continuity](#) to the [sequence stratigraphic framework](#) and the [biostratigraphy](#) provides the time constraints. All these different [sequence](#) stratigraphic techniques can be used independently of each other to produce accurate interpretations of the depositional histories of the sedimentary fill of a [basin](#) but the best models come from a mix of all three.

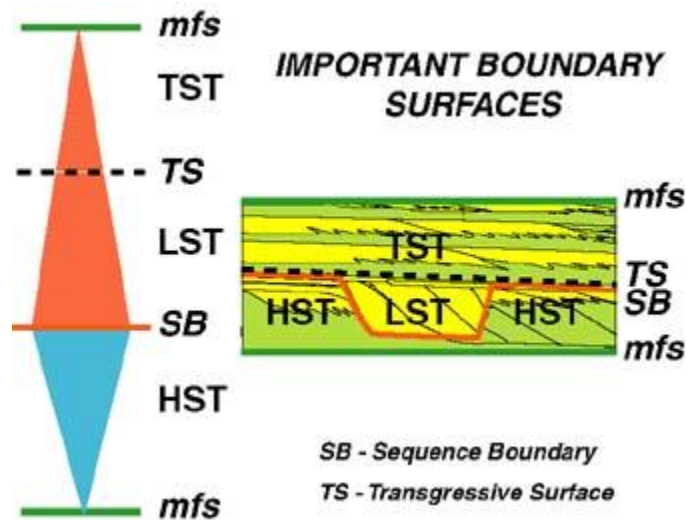
How well logs are used to interpret [sequence stratigraphy](#)

In the use of [sequence stratigraphy](#) the interpretations are best determined when well logs are tied to biostratigraphic markers. Using these two in combination one can:

- Identify, match and tie [sequence stratigraphic surfaces](#)
- Interpret the stacking patterns of the vertical sedimentary [sequences](#)

The character of electric logs of wells that penetrate clastics often reflect changes in grains size and so are easier to use in this process, while the logs of wells that penetrate [carbonates](#) often should be calibrated with cores, since [carbonates](#) are more susceptible to diagenesis and their change in character may be affected by more than

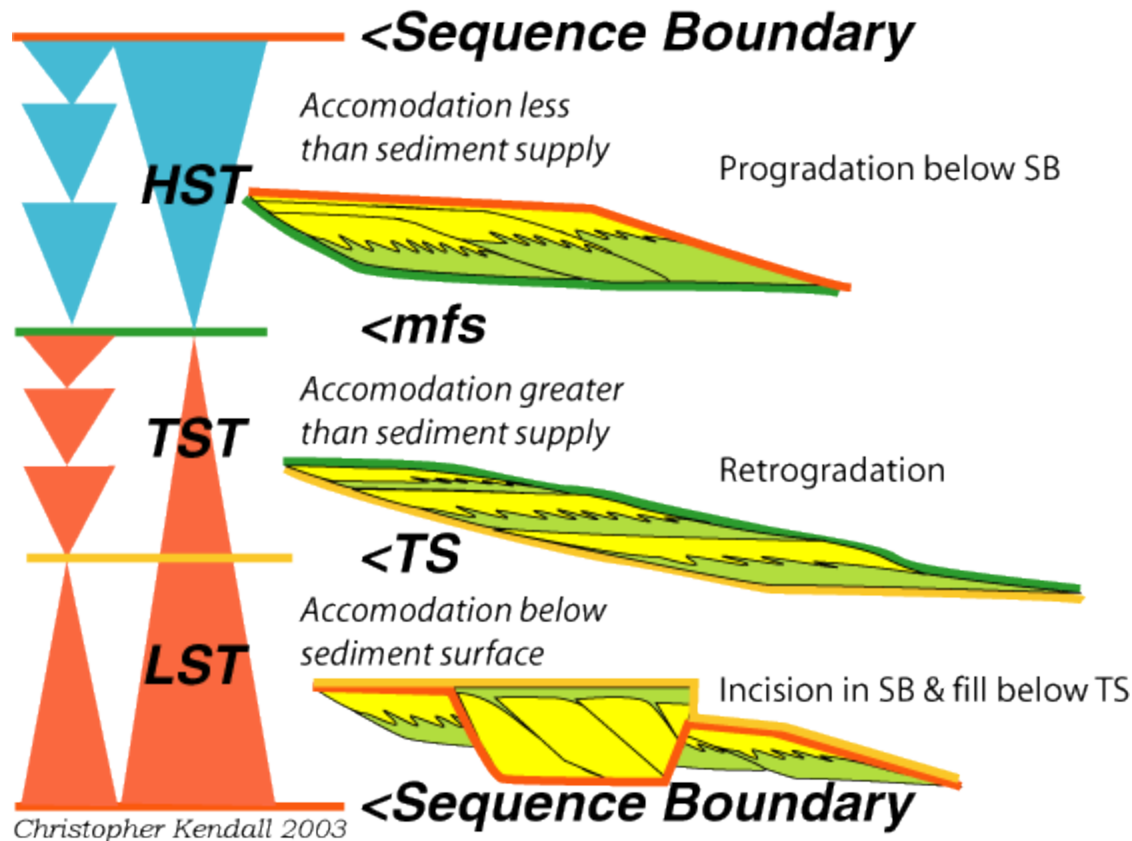
changes in [grain size](#). The sections that follow initially focus on the well log response to shallow water clastics and then move on to their response to shallow water [carbonates](#).



At the start of an interpretation of [sequence stratigraphy](#) using well logs one must first identify the predominant of [sequence stratigraphic surfaces](#). The MOST important of these [surfaces](#), and the FIRST that should be identified when using logs, are [maximum flooding surfaces](#) (mfs) and [transgressive surfaces](#) (TS). These coincide and are correlated with radioactive [shales](#) (use of the [gamma log](#)) that are interpreted to have been deposited across

relatively flat [surfaces](#). Once the mfs and TS are established and tied, then the [sequence boundaries](#) (SB) of both [carbonate](#) and clastic sedimentary systems are identified. These will tend to lie directly beneath the sand sized sediment fill of depressions on eroded and incised [surfaces](#) and over the prograding [clinoforms](#) of [high stand systems tracts](#) (HST).

Well logs used to interpret shallow water clastic [sequence stratigraphy](#)



In both clastics and [carbonates](#) the second and often co-incident step in the interpretation of well logs and cores is the use of [parasequence stacking patterns](#) (the vertical occurrence of repeated [cycles](#) of coarsening or fining upwards sediment) of to identify the [lowstand systems tracts](#) (LST), [parasequence cyclic stacking patterns](#) are commonly identified on the basis of variations in [grain size](#) and when these fine upwards are indicated by triangles whose apex is up while those that coarsen upwards are indicated by inverted triangles whose apex is down. The example presented in the above diagram catches the essence of the response of shallow water clastics responding to a fall and rise in [base level](#) and the incision and fill of the shelf margin during this [cycle](#).

The repeated [stacking patterns](#) for LST [cycles](#) are: -

- [cyclic](#) fill of incised depressions that tend to fine upward.
- [cyclic](#) sand to [shale](#) bodies of [basin](#) floor fans that tend to fine and **thin** upward.
- [cyclic](#) sand to [shale](#) bodies of shelf margin [clinoforms](#) that tend to coarsen and **thicken** upward.

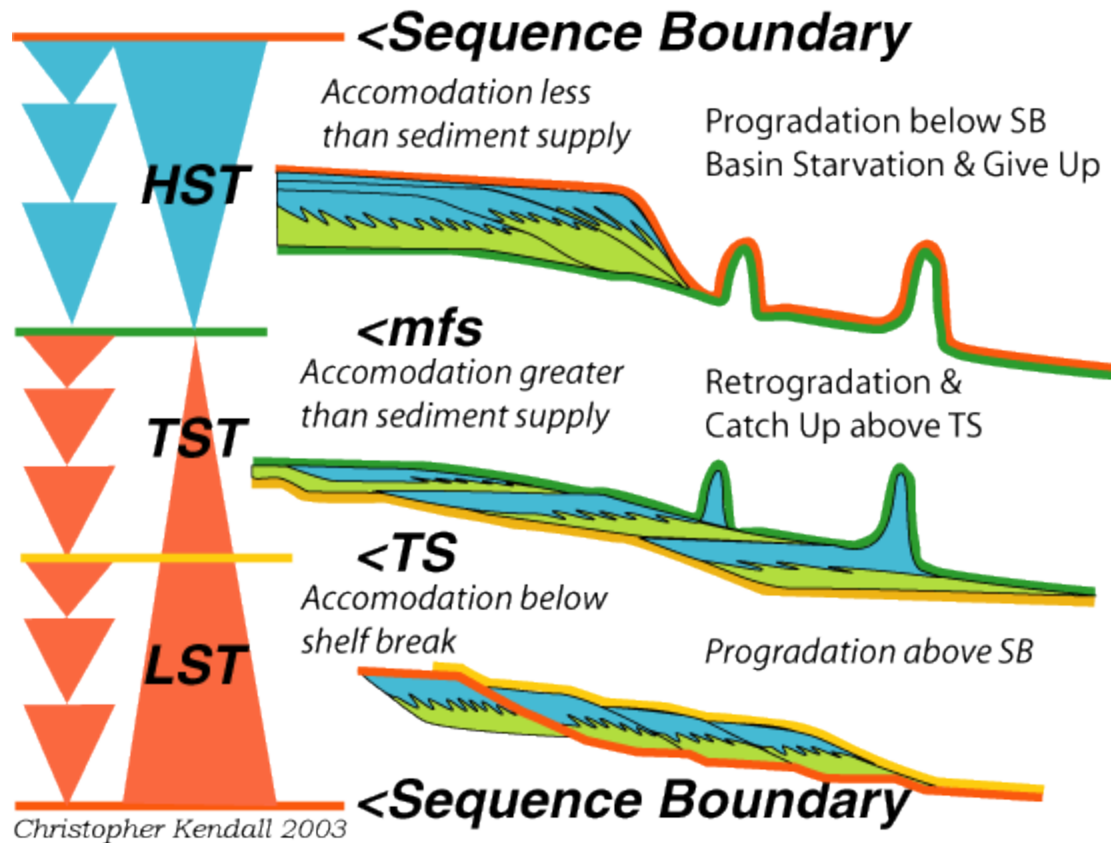
The repeated [stacking patterns](#) for TST [cycles](#) are: -

- Regressive [cyclic shale](#) to sand bodies of that tend to coarsen and **thin** upward.

You are advised to link to the exercises and movies on the clastic well logs of the Guarico [basin](#) on this site to learn more about these techniques.

Well logs used to interpret shallow water [carbonate sequence stratigraphy](#)

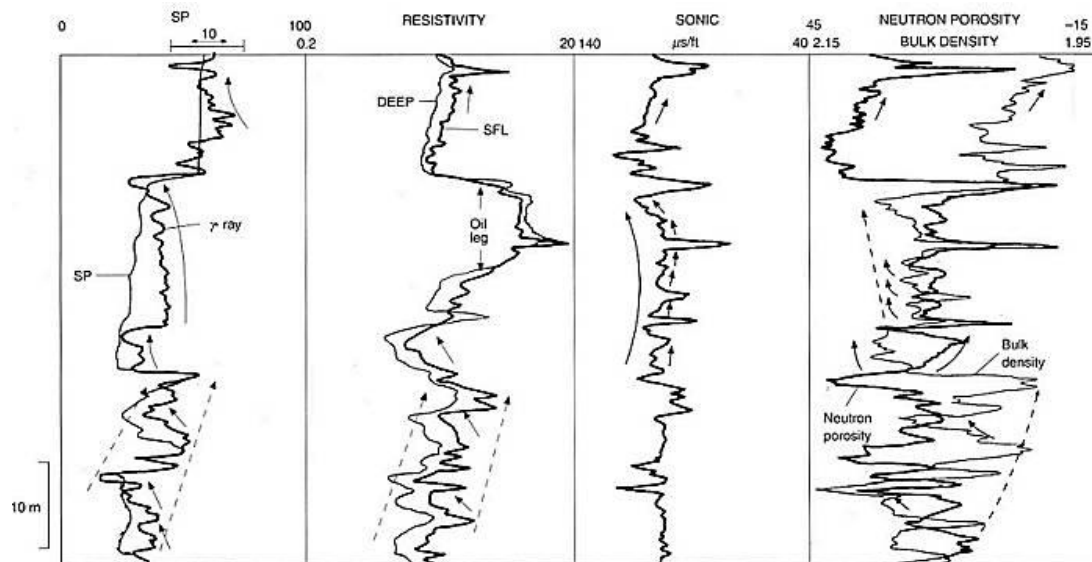
The above examples are focused on near shore shelf clastics. The same approach should be taken for [carbonates](#), with the second co-incident step in the interpretation of [carbonate](#) well logs and cores being to identify [parasequence stacking patterns](#) (the vertical occurrence of repeated [cycles](#) of coarsening or fining upwards sediment) of to identify the [lowstand systems tracts](#) (LST), [transgressive systems tracts](#) (TST) and [highstand systems tracts](#) (HST) that are enveloped by the [mfs](#), TS and SB. As in clastics, [carbonate parasequence](#) and [cycle stacking patterns](#) are commonly identified on the basis of variations in [grain size](#) and when these fine upwards are indicated by triangles whose apex is up while those that coarsen upwards are indicated by inverted triangles whose apex is down. [For more information on this topic you should click here to link to the carbonate sequence stratigraphy page of this site.](#)



Value of [sequence stratigraphy](#) built from well logs

Interpretations that use well logs for [sequence stratigraphy](#) should be tied to [biostratigraphy](#) so they can be used to correlate and analyze sedimentary [Rocks](#) from the perspective of geologic time. Well logs lend themselves to the detailed reconstructions of paleogeography and the generation of high frequency stratigraphic models that predict the distribution of sedimentary facies, particularly those associated with aquifers, sediment bound ore bodies, and hydrocarbon reservoirs, their source [Rocks](#) and seals.

[Well Log Interpretations](#) > [Intro Well Log Seq Strat](#) > [Well Log Suites](#)



(source: Emery, 1996)

Resistivity logs

This log measures the bulk resistivity (the reciprocal of conductivity) of the [formation](#). Resistivity is defined as the degree to which a substance resists the flow of electric current. Resistivity is a function of [porosity](#) and pore fluid in a rock. Porous rock containing conductive fluid (such as saline water) will have low resistivity. A non-porous rock or hydrocarbon-bearing [formation](#) has high resistivity. This log is very useful for determining the type of fluids in [formations](#) and is frequently used as an indicator of [formation lithology](#).

Spontaneous potential (SP) logs

This log measures the electrical current that occurs naturally in boreholes as a result of salinity differences between the [formation](#) water and the borehole mud filtrate ([formation](#) and surface). These logs are used as indicators of permeable [beds](#) (including determining permeable sands and impermeable [shales](#)) or for locating [bed boundaries](#). The SP log was one of the first tools to be used to distinguish [shale](#) from sand in clastic [sequences](#) (zero matches pure [shale](#) while high SP values match sand).

Gamma ray (GR) logs

This log records the radioactivity of a [formation](#). [Shales](#) (or [clay](#)-minerals) commonly have a relatively high gamma radioactive response, and consequently [gamma ray logs](#) are taken as good measures for [grain size](#) (and subsequently inferred depositional energy). Thus coarse-grain sand, which contains little mud, will have low gamma ray value, while a fine

mud will have a high gamma ray value. The values range of gamma ray is measured in API (American Petroleum Institute) units and range from very few units (in anhydrite) to over 200 API units in [shales](#).

[Gamma ray logs](#) are one of the most commonly used logs for [sequence](#) stratigraphic analysis. As explained above the variation in GR log character is interpreted to be a [proxy](#) for [grain size](#) and so enabled these logs to play a major role in the [sequence stratigraphy](#) analysis. The rationale behind this interpretation is that variations in the GR are related to the presence of organic matter that sequesters radioactive minerals responsible for the GR signal. In this case it is possible to: 1) identify the portions of the section which had higher [accumulations](#) of organics and 2) those for which this has been reduced by [winnowing](#) and oxidizing effects! In turn, the presence or absence of organic matter is assumed to be a [proxy](#) for the [winnowing](#) and oxidizing effects of shallow marine waters. The result is that abrupt changes in the GR log response are interpreted to be related to sharp lithological breaks associated with unconformities and [sequence boundaries](#). The principle shapes that GR log signal makes frequently used for interpreting the depositional setting of sedimentary [cycles](#).

[Neutron logs](#)

This log measures the [porosity](#) of a [formation](#), indicating in its response the quantity of hydrogen present in the [formation](#). The log is calibrated to [limestone](#). The linear [limestone porosity](#) units are calibrated using the API Neutron pit in 19% [porosity](#), water-filled [limestone](#) is defined as 1000 API units. This log is useful in measuring [lithology](#) (usually in combination with [density log](#)).

[Density logs](#)

This log is a measure of the [formation's](#) bulk density and is mostly used as a [porosity](#) measure. Different lithologies can also be determined using [density log](#) based on returned density value. For example, pure [quartz](#) will have a bulk density (g/cm³) up to 2.65, [coal](#) 1.2-1.8, [halite](#) 2.05, [limestone](#) up to 2.75, [dolomite](#) up to 2.87, anhydrite 2.98. Density is mostly commonly used in conjunction with [neutron logs](#) to determine [lithology](#) of [formation](#) (density-neutron suites such as Schlumberger FDC-CNL suite).

[Sonic \(acoustic\) Logs](#)

This log measures of the speed of sound in the [formation](#), and is related to both the [porosity](#) and [lithology](#) of the rock being measured. Thus, if the [lithology](#) of a [formation](#) is known, this log can be used to determine its [porosity](#). [shales](#) have lower velocity (higher

transit time) than sandstone of same [porosity](#), making this log a good indicator of [grain size](#). [sonic log](#) values (in ms/ft) for some rock types are: sandstone 51-56, [limestone](#) 47.5, [dolomite](#) 43.5, anhydrite 50, [halite](#) 67.

There are three general trends or curve shapes that can be recognized when looking at well log curves. For instance the [gamma log](#) tracks the upward change in [clay](#)-mineral content.

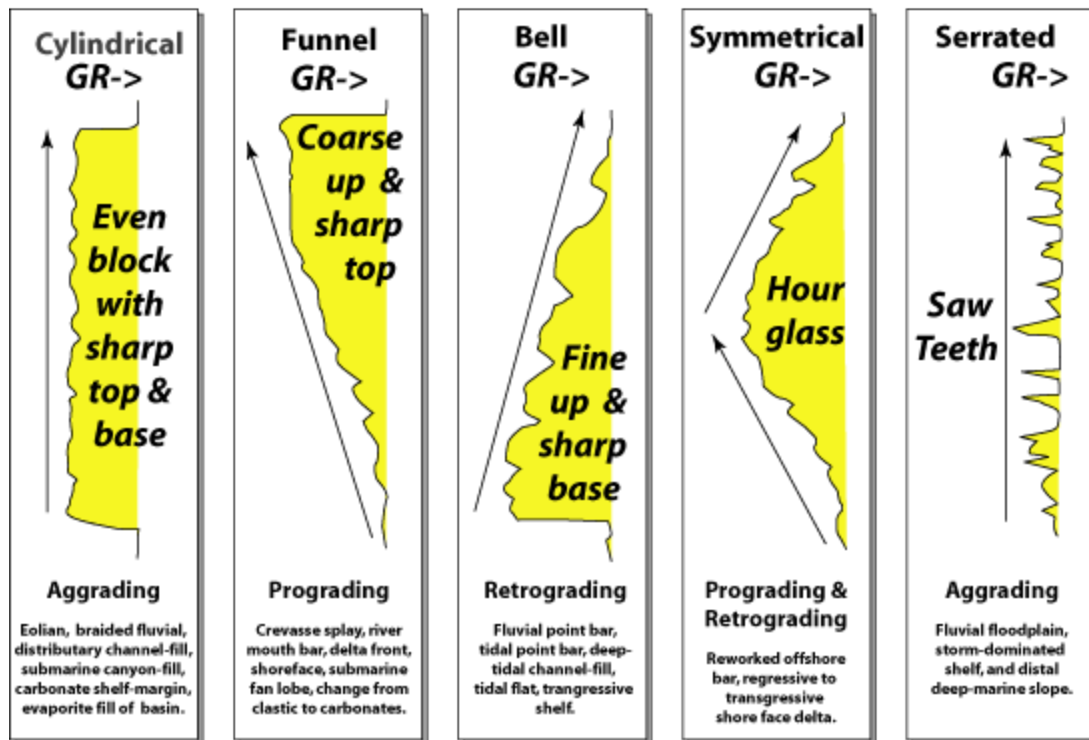
NB: whenever possible, one should use core data to confirm or augment well log analysis. This because core provides direct access to the character of the rock penetrated by the well whereas well logs are used to infer the character of the rocks penetrated. Care in establishing and matching the depths of the core and well logs is essential in the correlation of well logs and core.

Gamma Ray Response as a [proxy](#) for [grain size](#)

Variation in GR log character is interpreted to be a [proxy](#) for [grain size](#) and so enable these logs to play a major role in the [sequence stratigraphy](#) analysis. The rationale behind this interpretation is that variations in the GR are related to the presence of organic matter that sequesters radioactive minerals responsible for the GR signal. In this case it is possible to: 1) identify the portions of the section which had higher [accumulations](#) of organics and 2) those for which this has been reduced by [winnowing](#) and oxidizing effects! In turn, the presence or absence of organic matter is assumed to be a [proxy](#) for the [winnowing](#) and oxidizing effects of shallow marine waters.

The result is that abrupt changes in the GR log response were interpreted to be related to sharp lithological breaks associated with unconformities and [sequence boundaries](#) (Krassay, 1998). The principle GR log shapes were frequently used for interpreting the depositional setting of sedimentary [cycles](#).

General Gamma Ray Response to Variations in Grain Size



C. G. St. C. Kendall 2003 (modified from Emery, 1996)

- **Cleaning-up trend (funnel shape)**: a gradual upward decrease in gamma response. In shallow marine settings, this trend reflects a change from [shale](#)-rich into sand-rich [lithology](#) and upward increase in depositional energy with shallowing-upward and coarsening. In deep marine settings, this trend reflects an increase in the sand contents of [turbidite](#) bodies. This trend also may indicate gradual change from clastic to [carbonate](#) deposition.
- **Dirtying-up trend (bell shape)**: a gradual upward increase in gamma response: This trend may reflect upward fining (eg: a [lithology](#) change from sand to [shale](#)) or upward fining of sand [beds](#) in a thinly interbedded sand-[shale](#) unit. This trend usually implies a decrease in depositional energy. In a non-marine setting, fining upward is predominant within meandering or tidal channel deposits with an upward decrease in fluid velocity within a channel (coarser sediments at base of channel). In a shallow-marine setting, this trend usually reflects an upward deepening and a decrease in depositional energy ([shoreline](#) retreat). In deep-marine settings, this trend reflects waning of submarine fans (reducing of sand contents).
- **Boxcar trend (cylindrical or block shape)**: with low gamma and sharp [boundaries](#) and no internal change: this trend is predominant in fluvial

channel sands, [turbidites](#) (typically with greater range of thickness), and aeolian sands. [evaporites](#) also can have a cylindrical gamma trend.

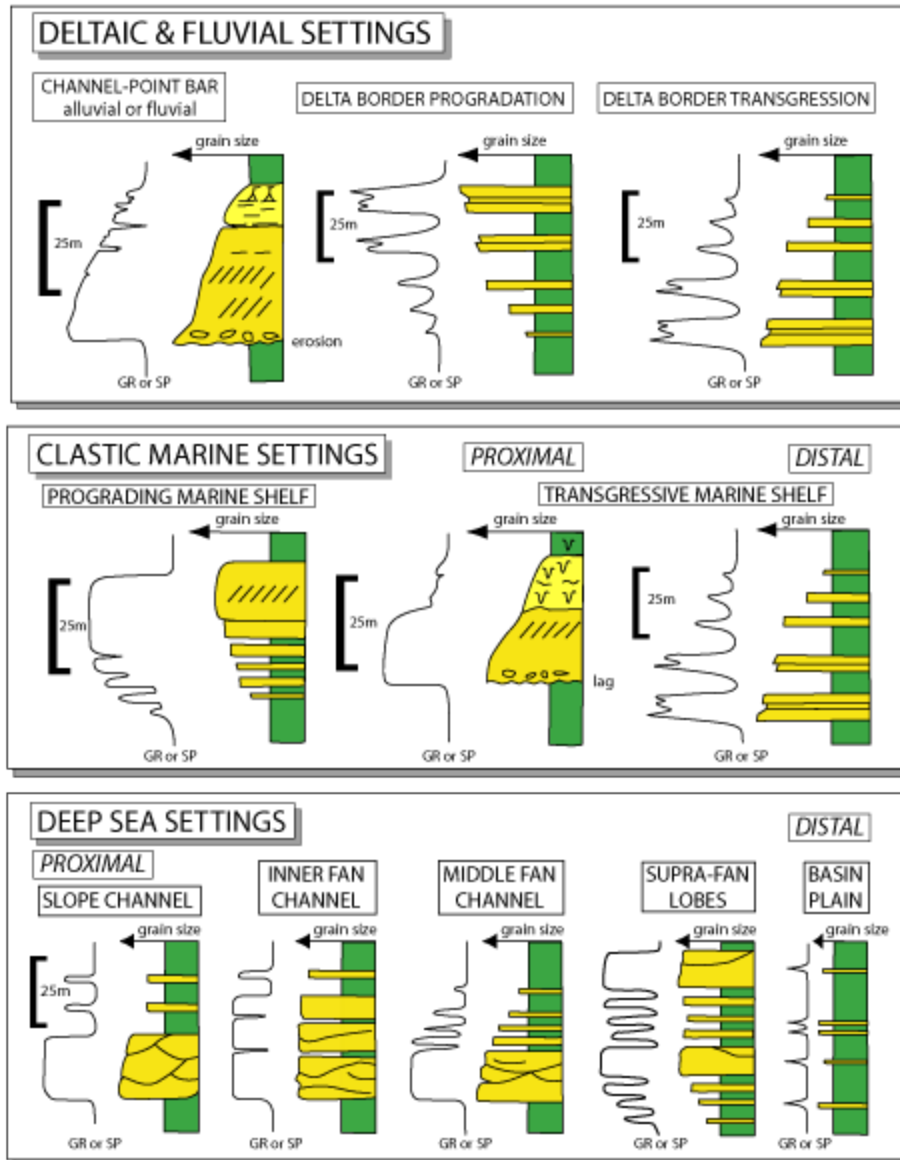
Additionally two in-between trends can be recognized:

- **Bow trend** (*symmetrical or barrel*): with gradual decrease then gradual increase in gamma response: this is usually the result of [progradation](#) and retrogradation of clastic sediments.
- **Irregular trend** (*lack of character*): this trend represents [aggradation](#) of [shales](#) or silts and can occur in other settings.

Curve character can be smooth, complex or serrated (sawlike) with contacts can be sudden or gradual.

The figure below summarizes the log response a variety of different clastic [depositional systems](#) that Malcolm Rider (1996) assembled in his excellent book on Well Logs. His diagram has been added to and slightly been modified for this web site.

Gamma Ray Log Response & Depositional Setting



C. G. St. C. Kendall 2003 (modified from Malcolm Rikter 1999)

[Sequence Stratigraphy](#) > [Well Log Interpretations](#) > [High Frequency Clastic Seq Str](#)



Introduction to well log character and [shoreline](#) clastics - Venezuela

Much of the un-referenced text below is copied and modified from John Reistroeffer's (2001) PhD Dissertation. Paul Lake, Eric Anderson and Nassir al Naji helped build the exercises from John Reistroeffer's data set. Errors are those of Christopher Kendall (alias "Retorno al Futuro") who ascribes to John's positive credo that "the sky is the limit"!

The exercises that follow introduce the interpretation of the [sequence stratigraphy](#) of an area using the electric logs. These logs are from a series of wells drilled in the [shoreline](#) clastics of the Mesozoic and Tertiary section of the northern [Cratonic](#) margin of Eastern Venezuela in the Western Guarico sub-[basin](#). The focus of the exercises are wells that penetrate the Lower Oligocene La Pascua Formation. Since this [formation](#) does not outcrop, its [lithology](#) and sedimentary character are determined from E-logs and cores. The La Pascua Formation is an overall [transgressive](#) sandstone unit that contains the major hydrocarbon reservoirs for the region which have produced some 135 million barrels of oil and 686 billion cubic feet of gas (Reistroeffer, 2001). The exercises use electric logs to first identify and correlate a series of [stratigraphic surfaces](#). These include [transgressive surfaces](#)-TS, [maximum flooding surfaces](#)-MFS and [sequence boundaries](#)-SB. From these the [stacking patterns](#) of the enclosed [parasequences](#) are established and used to determine the depositional setting of this [formation](#).

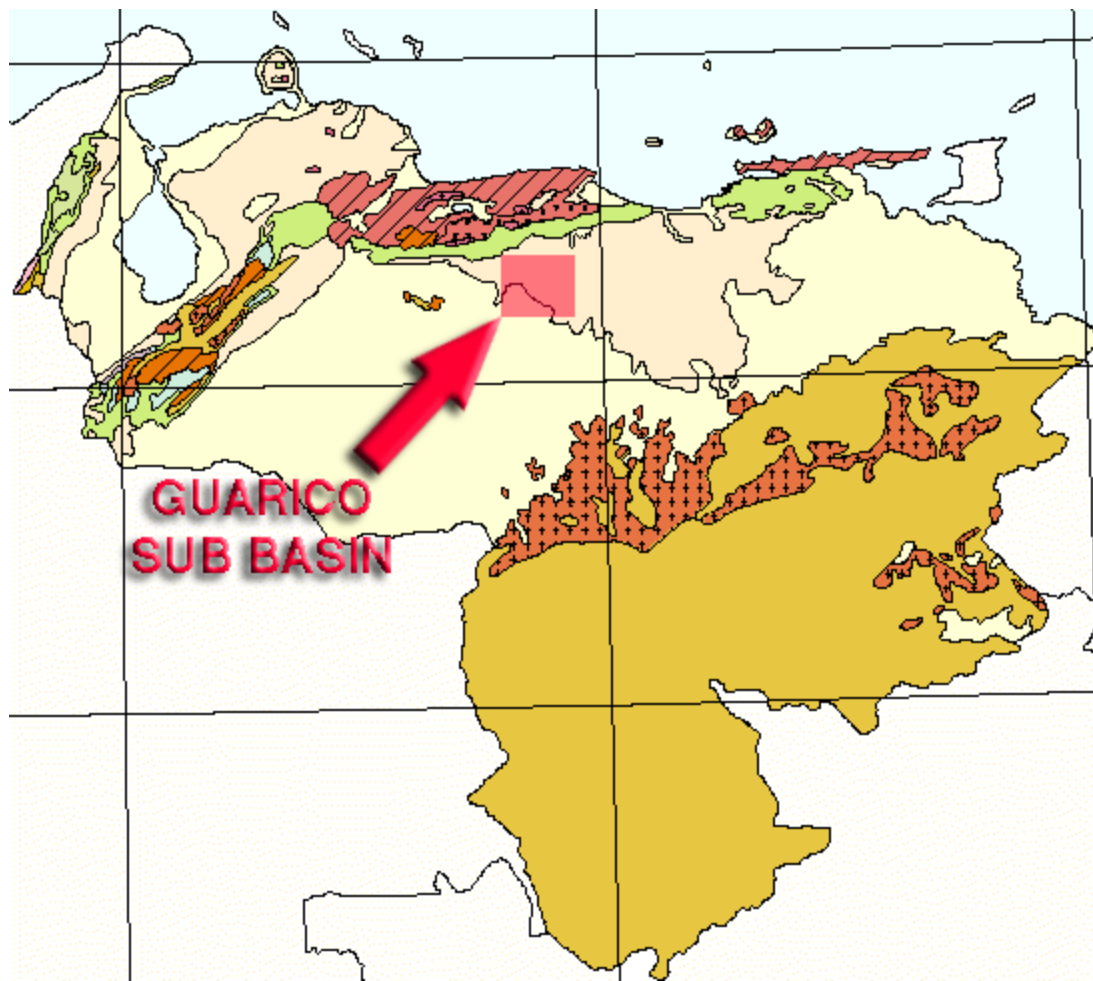


Figure 1: Generalized geologic map of Venezuela showing the location of the Western Guarico block and the chosen study area.

(www.geology.about.com/library/bl/maps/blvenezuelamap.htm)

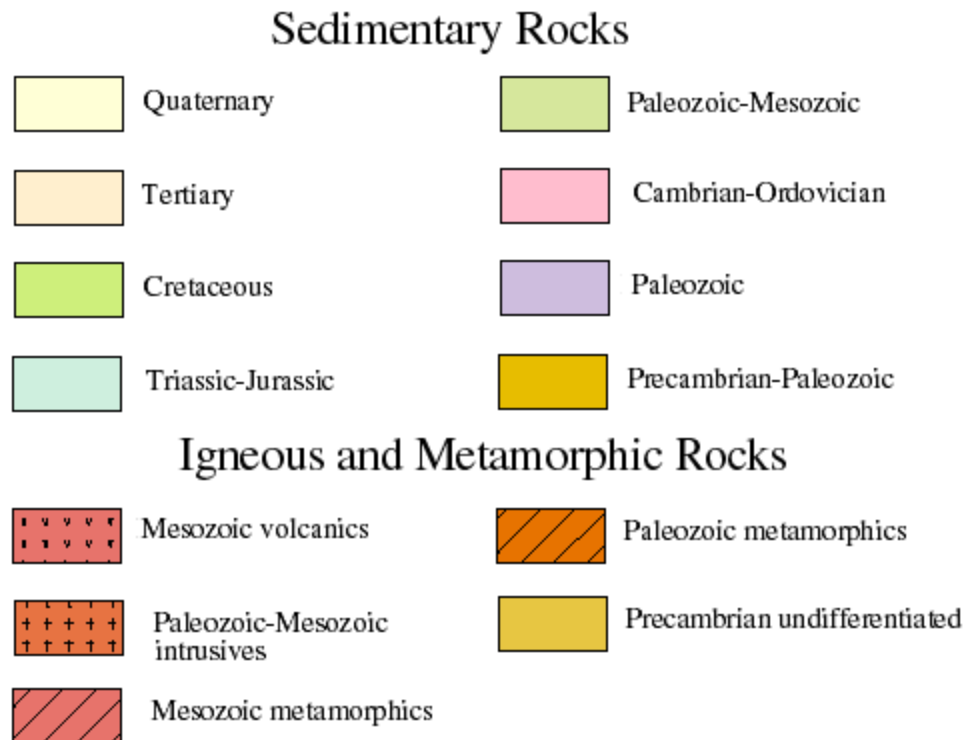
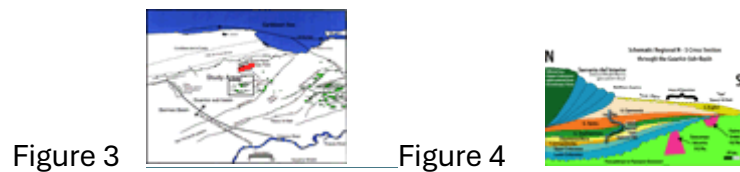


Figure 2. Key to the generalized geologic map of Venezuela (figure 1 above)
 (www.geology.about.com/library/bl/maps/blvenezuelamap.htm)

The Eastern Venezuela Basin is formed of a series of Late Eocene to recent foreland basins located between the Guayana Shield to the South, and mountain chains associated with the Caribbean-South American plate to the north. The Caribbean-South American plate margin became active in the late Cretaceous when the Caribbean plate moved eastwards along the northwest margin of South America (Burke et. al 1984, Lugo, 1994, Audemard et. Al. 1985, Bartok, 1993). This eastward movement relative to and past the South American plate resulted in compressional to transpressional deformation along the interface between the plates. The thickest sedimentary packages of these asymmetric basins are located in the domain of maximum subsidence along an axis parallel and adjacent to the mountain front. Sediments thin in a southerly direction away from this axis. The Western Guarico sub- basin, the site of the well log exercises, is one of these basins and the basin surface changes into a gently dipping zone of low subsidence and sedimentation rates and the Craton forms the southern margin of the basin dips north towards the basin axis.

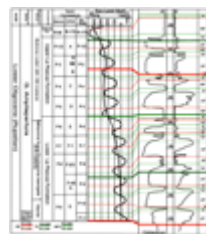
The sediments of the La Pascua Formation accumulated during a transgression in the

Tertiary, and [onlap](#) onto the underlying Cretaceous Tigre Formation. The area of study is part of the Western Guarico Block (Figures 1, 2, 3 and 4).



La Pascua Formation - Origin of the Oligocene [cycles](#)

Reistroeff (2001) described the La Pascua section used in these exercises as corresponding to a third order Lower Oligocene [sequence](#) which lasted slightly more than one million years. He believed that this [sequence](#) is composed of ten fourth order [sequences](#) that each lasted about 125 thousand years. He noted how high frequency Oligocene [sequences](#) is also found to the east in the same age sediments of the Los Jabillos Formation (Personal Communication to Reistroeff from Gerald Baum). In the east in the northern Monagas area Genaro Gifuni (1996) identified low and high frequency [sequences](#) in the Upper Oligocene Naricual Formation. The Rupelian aged Boom [clay](#) of Belgium is also composed of [Rocks](#) that are interpreted to have accumulated over a series of 100,000 year [cycles](#) and Abreu and Haddad's (1998) unsmoothed oxygen [isotope](#) curve has high frequency "noise" with a similar [cyclicality](#). These latter authors attribute high frequency [cyclicality](#) of the Boom [clay](#) to Milankovich [cycles](#) and attribute the driver to something similar to the third order [cyclicality](#) seen in the Antarctic deep sea drilling cores glacial [cycles](#). Hanners (1974) also attributed la Pascua Formation [cyclicality](#) to [Milankovitch cycles](#). Thus together the thread that ties these different Oligocene sections together is interpreted by Reistroeff (2001) to be the fourth order [cycles](#), attributed to [Milankovitch](#) driven glacial [cycles](#), and their superposition on third order [cycles](#)



as in Figure 5.

La Pascua Formation - General Description

As indicated above the La Pascua Formation is confined to the subsurface. It was first formally described by Hedberg (1950) but was informally used by the SAPLM in their early exploration of the Las Mercedes area. It was later described in detail by Patterson and Wilson (1953) and they designated the type section as the Guayabo-2 well in the southeastern quadrant of the study area.

The La Pascua Formation is approximately 500 feet thick in the Las Mercedes field area. Approximately 300 feet of this is net sand, and 200 feet is [shale](#). Eight principle sands were labeled by Sociedad Anonima Petrolera Las Mercedes (SAPLM) from youngest to oldest as "A", "B", "BB", "C", "D", "E", "F", and "G" (Figure 5 thumbnail above). This correlation was changed by John Desisto working for Meneven in 1977 to be nine sand bodies ("P-1" through "P-9") from youngest to oldest. The exercises below Reistroeffers 2001 nomenclature, expanding on that of Desisto by identifying eleven sand units that included P-1 through P-11 (Figure 5 thumbnail above). The difference is reflected with two additional sands in SAPLM's "F" series, and an additional [transgressive](#) unit which pinches out at the level of the lower "G" sand.

The sands of the La Pascua Formation are grey to grey-brown and range in average [grain size](#) from very fine to very coarse with some conglomeratic intervals containing cobble-sized material (Patterson and Wilson, 1953). Total silt and [clay](#) content in the better developed sands varies from 4.6-13% (Perko, 1952). Overall the sands have fair to good sorting. Sieve analysis shows a tendency towards finer [grain size](#) at and above the level of the "D" sands (P-5) (Perko, 1952). [kaolinite](#) is often present in the sands ranging from point occurrences to thin layers. Often the fines are associated with production problems. This material is interpreted to be weathered remnants of feldspars derived from [granites](#) of the Guyana shield and the El Baul arch.

The [shale](#) layers that separate the sandstone bodies are black to grey-black, massive to finely [laminated](#), calcareous to non-calcareous, with occasional [pyrite](#) grains and thin layers of lignite. Lignites and lignitic [shales](#) are common in the section, and, where they have been cored, reach four feet in thickness. These are commonly found near the top of each sandstone interval. Typical fossils include mollusk and mollusk fragments, which are often found in the sand bodies. Plant impressions and remains are commonly found within the levels of the lignite and lignitic [shales](#). Foraminifera are found in the [shales](#) between the sand bodies (Patterson and Wilson, 1953).

The heavy minerals of the La Pascua Formation are dominated by titanium minerals of which leucocene (weathered illmenite) usually makes up more than half of the heavy concentrates, along with smaller amounts of anatase, rutile, titanite, and brookite. The non-titanium minerals are dominated by zircon, with scattered occurrences of tourmaline. Scattered single-grain occurrences of [staurolite](#), hornblende, augite, [epidote](#) and hypersthene are also present (Perko, 1952). The La Pascua Formation sands in the West Guarico block have a similar heavy mineral suite to the Merecure Group sands, as

discussed by Funkhouser et. al. (1948) as having been derived from the Guyana Shield.

The La Pascua Formation is overlain by the regional [transgressive](#) Roblecito Formation, a marine facies which has a [proximal](#) La Pascua sandy equivalent. The original definition of the La Pascua Formation only included the sandy intervals found in the Las Mercedes field area. However this name has come to represent the basal sandstone units of the regional Upper Eocene-Lower Oligocene [transgression](#). The sandstone levels at and above the "D" sands (P-5) in the Western Guarico block are represented by [shales](#) to the northwest in the Grico, Camaz, and El Sombrero areas. Sandstone levels in the north and northwest are older, while successively younger units pinchout southeastward. Basal sandstones to the south, in the Machete area of the heavy oil belt, are the lateral equivalents to marine [shales](#) of the Roblecito Formation, which overlie the Western Guarico Block.

Sandstones in the La Pascua Formation are derived from three primary sources. The lower units ("G" and "F" SAPLM nomenclature, or P-7, P-8, P-9 Desisto nomenclature or P-7, P-8, P-9, P-10, P-11, these exercises) are all derived from a southern source in the Precambrian Roraima Formation of the Guyana shield south of the area of the exercises. Sands in the upper La Pascua Formation at and above the "D" sands (the P-5 sands of Desisto and these exercises) in the Western Guarico Block are also derived from the southern Guyana shield source. Upper La Pascua Formation sands north of the study area in Yucal-Placer, Macaira, Uveral and Cardonal are derived from a northern source, probably eroded material from an encroaching northern thrust front (Aymard et. al., 1985, Daal et. al. 1989). Between the area of the exercises and the above-mentioned areas to the north, exists a zone with minimum sandstone present in the La Pascua Formation. This area is the axis of the La Pascua aged [basin](#), which separates the zone of southern-sourced clastics from the zone of northern-sourced sediments (Patterson and Wilson et. al. 1953, Fasola et. al., 1985 and Cabrera et. al., 1994). In addition, the presence of [kaolinite](#) in the sands opens the possibility that at least some of the clastics are derived from the El Baul arch.

The upper contact with the Roblecito Formation is transitional and conformable. It corresponds to the top of the uppermost sand-body in the Guayabo-2 well in the study area. The contact in the northwest which corresponds to an electric log marker at the base of the Roblecito [shales](#), does not represent a lithologic change, since the upper La Pascua section is mostly [shale](#). Likewise; to the northeast of the study area, the contact is marked by an electric log marker at the top of the sandy La Pascua Formation interval but does not represent a lithologic break, since the Lower Roblecito Formation is also quite sandy.

The Lower contact with the Cretaceous El Tigre Formation is unconformable, and the La

Pascua sands overlie progressively older sediments in a southwesterly direction until they rest unconformably on the Paleozoics of the Barbasco Group, the El Baul [granite](#) and the Guacamaya Volcanics, southwest of the town of El Calabozo (Miller and Martinez, 1972, Patterson and Wilson, 1953).

The contact between the basal La Pascua and the Cretaceous was extensively cored by SAPLM. The [unconformity](#) is easy to determine in cored sections but is often less obvious from cuttings and E-logs. Cores show poorly-sorted sandstones and [conglomerates](#) ranging in [grain size](#) from fine to pebble sized. Geode-like silica concretions, chert grains and pebbles considered to be reworked Cretaceous sediments are common. [kaolinite](#) occurs in small pockets which represent the weathered remains of feldspar crystals. Occasional lignites and lignite streaks are found in the lower La Pascua sandstone. Dr. L. V. Illing et. al. in a 1953 internal report describes the pebbles in the basal Tertiary unit as follows:

"... well rounded all sizes up to 3" across: the dominant size is estimated at one half to three quarters inch in diameter. The principle (pebble type) white porcelaneous chert with a slightly powdery texture and often with roughly cuboidal fractures. Some pebbles show a darker core. The white color is due to a very finely porous texture, and is clearly due to alteration, presumably weathering. By soaking in clove oil, the chert becomes translucent or transparent, and shows small inclusions and occasional foraminifera. Some pebbles are very rich in foraminifera, including abundant Textularia(?), some Globigerina and many other pelagic forms. The chert is devitrified and shows as a plexus of tiny crystallites under cross-nicols. Occasionally, it shows a spherulitic devitrification. Rare pebbles of fresh chert occur"

The Underlying Cretaceous has blue-grey fish remains, rodlike tan fecal [pellets](#), foraminiferal [chert](#), isolated secondary [dolomite](#) rhombs and finely banded siliceous argillite. Occurrences of black translucent chert have abundant inclusions of foraminifera including spherical cells, Textularia and Globigerina. Occasional veins of opaque white and clear blue chert occur, the white color is probably due to weathering similar to the weathering that occurs in the La Pascua Formation cores (Illing et. al., 1953).

The La Pascua Formation, in wells adjacent to the southern margin of the mountain front in the northern Guarico area, is Lower Oligocene in age, based on the presence of the planktonic foraminifera zones of Cassigerinella chipolensis/Pseudohastigerina micra and Globigerina ampliapertura (Fasola et. al., 1985).

The next **Section** describes the first steps in the use of well logs to build [sequences](#) stratigraphic models of clastic [shoreline depositional systems](#) for the Lower Oligocene La Pascua Formation of the Las Mercedes Field in the West Guarico Block, Venezuela.

Introduction to exercises in well log response of [shoreline](#) clastics - La Pascua [Formation](#), Venezuela:

The exercises below involve detailed correlations using SP and Resistivity Logs. There two sets of these exercises and both are related to the interpretation of the local [stratigraphy](#) of the Lower Oligocene La Pascua [formation](#) of the Las Mercedes Field in the West Guarico Block of north eastern Venezuelan in which clastic [sequences](#) related to [incised valley](#) fill and their relationship to sea level.

The first set are [Exercise 1](#), [Exercise 2](#) and [Exercise 3](#) which cover a first pass with a well log data set from a limited number of wells for the top of the La Pasqua [formation](#) involving three [sequences](#). These well logs are flattened on the top of a major transgressive surface (TS) on the silt marking the contact between the La Pascua below and the overlying Roblecito [Formation](#).

The second set are [Exercise 4](#), and [Exercise 5](#) and these cover a well log data set also focuses on the interpretation of the top of the La Pasqua [formation](#) but now extends to five [sequences](#) and many more wells. To help with the interpretation of these latter logs an initial correlation has been made with the sections hung on three widespread [surfaces](#). These are:

- A basal surface correlated to the top of the P-5 silt
- A surface representing the top of the P-3 silt at the center of the section and around which it is flattened
- The top of the section is marked by the top of the silt that lies at the contact between the Roblecito and Pascua [Formations](#).

The correlation [surfaces](#) used in both sets of exercises extend across the area penetrated by the wells, except where they are interrupted by the incision of local channels.

These [surfaces](#) cap silty horizons that are equated with surfaces of transgression (TS) that formed at [wave base](#), and above, when the sea floor was reworked just following a sea level low. Normally radioactive peaks on Gamma Ray Logs, associated with mfs, are more extensive and would have been better [surfaces](#) but [gamma ray logs](#) were not available for much of the Guarico Sub [basin](#).

The strategy recommended for the exercises is to first identify additional [surfaces](#) capping through-going [shales](#) and silts and correlate these from well to well. Once

the [surfaces](#) have been identified these should be used to separate and correlate parasequence sets (at the scale of feet to tens of feet). The vertical and lateral character of these [parasequences](#) (stacking patterns) is in turn used to interpret the depositional setting of the component systems tracts. For instance these [boundaries](#) should be used to bracket the different [parasequence](#) intervals (as in the cross section displayed in the oval image heading this section) and then these [parasequences](#) are used to establish the [sequence stratigraphy](#) of the lower Oligocene La Pascua [formation](#) in the West Guarico Block.

In the exercises it is assumed that [parasequences](#) identified from the log character match universal and easily identifiable high frequency signals that can be used to correlate the marine and marginal marine sediments involved, particularly since the planktonic or palynologic biostratigraphic markers occur in only a few vertically dispersed [shale](#) horizons (Reistroeffer, 2001). The resulting [parasequences](#) have the same scale as the components of local hydrocarbon reservoirs. The [parasequences](#) have been interpreted, mapped and tied to high frequency [eustatic events](#) and then be used to find and exploit these reservoirs (Reistroeffer, 2001). Interestingly the Rupelian aged section of the exercises has more stratigraphic sequences (Vail et al., 1977) than have been recorded on the eustatic charts of Haq, et. al. (1987). This is explained to be the product of high frequency variations in one or more of the principal variables that controlled the dimensions of the accommodation space. These variables include rates of subsidence (tectonics), sedimentation, and/or sea level. The economic impact of this high number of stratigraphic [sequences](#) is to increase the number of reservoir-seal couplets.

Criteria to use in making a depositional interpretation - a Big Secret!!!

The purpose in providing a background on the regional geology of the Guarico Sub [basin](#) was to help extend the data of the wells. While the scientific method encourages the scientist to gather data, form a hypothesis and then test this with further observation, it should be recognized that only **some** of the data available is relevant and essential to determining the depositional setting of these rocks and their distribution. The experienced interpreter filters the noise from the data, looks for criteria that are considered essential to solving the problem at hand, proposes a geological model, and checks to see if the data matches that expected for that model.

What are these magical essential criteria? They are often **uncited** since they are considered to be **too obvious**. These criteria vary from one geological problem to the next but in the interpretation of the [sequence](#) stratigraphy of an area they will tend to be very much related to the regional setting of the rocks in question. The experienced interpreter will either already know this, or will read the relevant literature to find out what has already

been established concerning the depositional setting of the sedimentary rocks in question. This understanding of regional geology will then be combined with a knowledge of vertical and lateral facies relationships in near shore clastic settings (eg. shoreline, beach, stacked beaches, tidal flats, deltas) and Walther's Law, and used to build potential depositional models. The interpreter will then work with the logs using techniques similar to those explained in the exercise to establish the correct depositional model and a [sequence](#) stratigraphic interpretation should follow!

A proposed depositional model

As is the suggested strategy for the implementation of the exercises Reistroeffer (2001) first tied the major [shale](#) markers and used well log character to interpret the distribution of the principal sandbodies of La Pascua [formation](#). He established these to be:

- Strike oriented wave-dominated detached-to-semi-detached high energy shoreface systems and barrier-island systems.
- Flood and ebb tidal delta, and tidal channel systems.
- Elongate dip-oriented tidal channel sands.
- [incised valleys](#) fill sands and massive-blocky, sharp-based widespread sands.

Sedimentary response to [base level](#) change and the corresponding log character

The vertical relationship of these sands is delineated on the figure below. Click on the thumbnails to view the depositional maps Reistroeffer (2001) created.



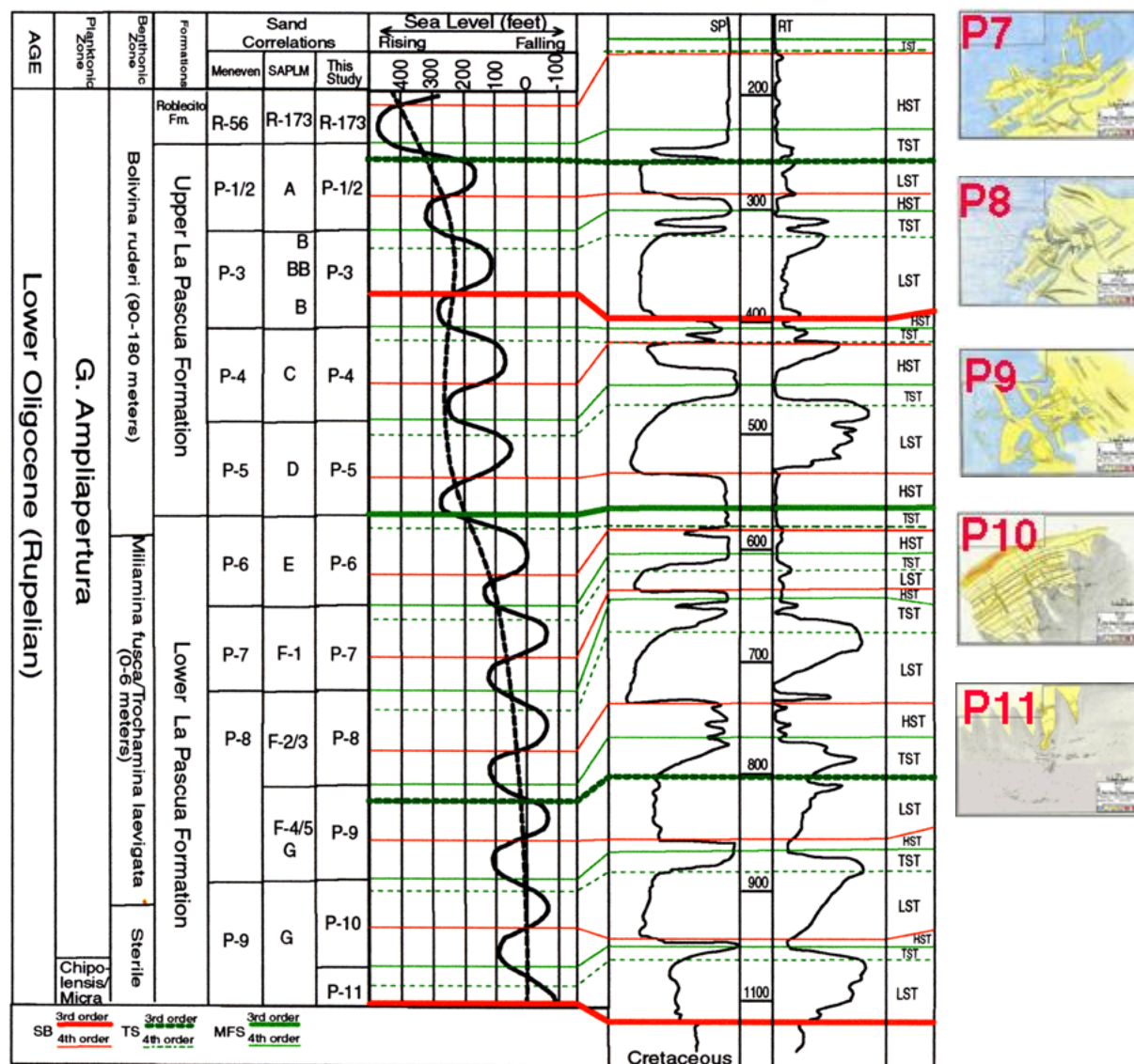


Diagram and maps of corresponding depositional settings by Reistroeffer (2001).

Reistroeffer (2001) examined the character and stacking patterns of the E-log response to establish the origin of each of these sand bodies, or groups of sand bodies, relating them to particular rates of relative sea level rise, fall or stillstand. Reistroeffer (2001) found that the higher frequency fourth order eustatic cycles were superimposed on a third order cycles of sea level. As shown on the Figure above he found that the dimensions of particular sand bodies could be equated with their location on the third order lower frequency sea level curve.